Martin Lau

Final Year X-HEC Data Science & AI for Business Master's student
Graduate in mathematics & computer science, trained data scientist and AI engineer.
Seeking end of studies internship in Data Science & AI starting April 2027.

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Experience

- **CHANEL SAS** *Apr 2026 - Aug 2026*
  AI Engineer Intern Paris, France
  - Improved the reliability of a LangChain-based agentic RAG assistant for boutique and Travel Retail teams by refining prompts, retrieval filters, tool logic, and output guardrails. Helped raise human-validated accuracy from 65% to 92%.
  - Built a multilingual BERTopic pipeline to transform 1,500 CHANEL boutique Google Maps reviews into actionable customer-feedback topics, integrating sentiment/emotion analysis, multi-label classification, and human/LLM validation. Communicated findings with several boutique owners through a Streamlit dashboard to improve client satisfaction scores.
  - Built a Machine Learning pipeline on Azure Maching Learning to predict customer return buyer scores based on historical behavior data. Handled SQL database ingestion, data preparation, feature transformation and evaluation metrics.

- **Fluke Corporation** *Jul 2025 - Aug 2025*
  AI Engineer Intern Shanghai, China
  - Led full deployment lifecycle, collaborating with backend/frontend teams to integrate AI features into Fluke flagship software.
  - Built NLP-to-SQL model using T5 transformers enabling non-technical users to perform complex database queries, eliminating SQL knowledge requirements.
  - Engineered custom dataset pipeline using Ollama, generating 100,000+ NL/SQL pairs across 7 languages.
  - Optimized inference speeds using quantization and hyperparameter tuning, reaching 97% extraction accuracy and reducing inference latency by 40% in real world tests.

- **Expando World Limited** *Jul 2024 - Aug 2024*
  Data Intelligence Intern Hong Kong SAR
  - Developed automated data collection engine for government tenders, processing 100+ documents per hour with multi-source integration and filtering capabilities to analyze government spending trends.
  - Designed and implemented scalable ETL pipeline for B2G sales intelligence, reducing manual data processing time by 90%.

Projects & Research

- **Bachelor Research Thesis** *Jan 2025 - Mar 2025*
  - *Data Stream Learning for Behavioral Cloning*, written under Professor Jesse Read (ORAILIX group, École Polytechnique).
  - Conducted a first of a kind review of the intersections between Reinforcement Learning and Data Stream learning, using TensorFlow and StableBaselines to reach 1.5x faster learning speed and 94% accuracy on simple problems.

Education

- **École Polytechnique, HEC Paris** *Sep 2025 - Jun 2027*
  Master of Science: Data Science & AI for Business Paris, France
  - Coursework: Statistics, Python for Data Science, Machine Learning, Database Management, Linear Algebra & Regression

- **École Polytechnique** *Sep 2022 - Jun 2025*
  Bachelor of Science (Mathematics & Computer Science) Paris, France
  - Graduated Cum Laude (GPA 3.84/4.00).
  - Relevant coursework: Machine Learning, Computer Networks, Distributed and Concurrent programming, OOP with C++, Linear Algebra, Real Analysis, Measure Theory and Integration, Probability & Statistics, Stochastic Processes.
  - Exchange semester at Shanghai Jiao Tong University (Sep 2024 - Jan 2025): Cloud Computing, Automation Engineering.

Skills

- **Programming Languages**: Python (NumPy, Pandas), Jupyter Notebook, C, C++, R, SQL, HTML/CSS/JavaScript.
- **ML Frameworks**: Scikit-learn, PyTorch, TensorFlow, Keras, HuggingFace Transformers, StableBaselines, LangChain.
- **Development tools**: Linux, Excel, Macros & VBA, Github, Docker & Kubernetes, Microsoft Azure Cloud.
- **Languages**: English (Native), French (Native), Mandarin (B2), Spanish (B1).

Extra Curriculars

- **Student Activities**: Creator and President of the Bachelor Cooking committee (2023), organized student events for over 50 people, including cooking workshops, charity bake sales, international buffets and cooking competitions.
- **Part Time Work**: Tutoring young students (middle & high school students) and coaching for university admissions.
- **Hobbies**: Sports (Tennis, Table Tennis, Badminton, Volleyball), Photography, Writing, Cooking.